

Physics 0525: Lab 5

Diode Circuits and Rectifiers

Equipment: Breadboard, digital multimeter (DMM), oscilloscope, 6.3 V transformer and/or signal generator, 1N914 rectifier diode (4), capacitors, resistors

Preliminaries:

A diode is made by joining doped p-type and n-type semiconductors together (a *pn* junction). It is a passive non-linear device with a current-voltage (I-V) curve shown on Figure 1. Because the current increases exponentially with voltage in the forward direction and the current is so small in the reverse direction, we can approximate the behavior of the diode as a one-way conductor:

(1) When the diode is biased in the forward direction (i.e. the anode potential is higher than the cathode potential) and the voltage difference is larger than the turn-on voltage ($V_{on} \approx 0.6$ V for Si diodes), the diode is conducting, its resistance becomes practically zero and the voltage drop across the diode is approximately equal to the turn-on voltage ($\Delta V \approx V_{on}$). In this case the current through the diode is limited by other components, typically a resistor, in series.

(2) When the bias voltage is lower than the turn-on voltage of the diode or it is reversed (anode potential is lower than the cathode potential), the diode resistance becomes very large and the current through the diode becomes practically zero ($I \approx 0$). The diode behaves like an opening in the circuit.

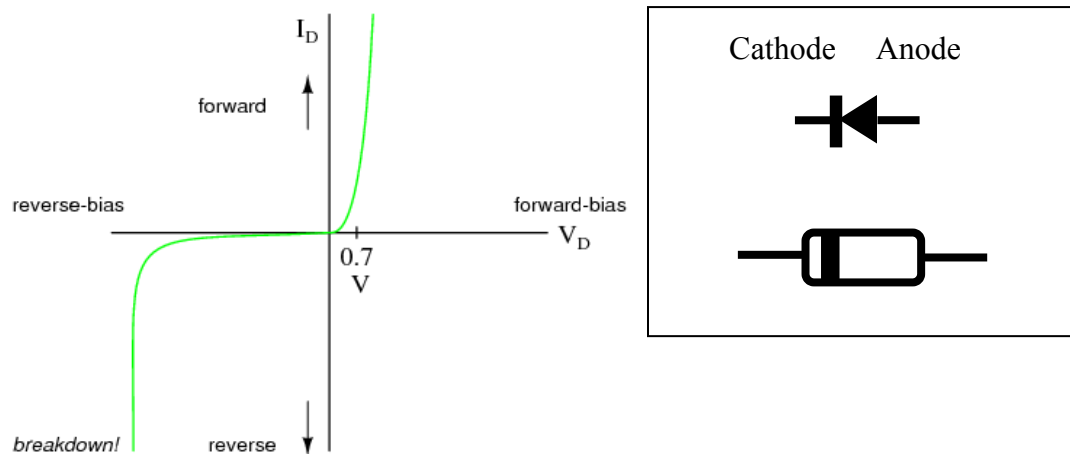


Figure 1. A characteristic I-V curve of a diode on the left. The circuit symbol and a simplified drawing of diode on the right. Note that the arrow representing the anode terminal points in the direction of the forward current.

Name: _____

Date: _____

Before you start working, identify the cathode and anode leads of the diode (see Fig. 1). Measure the forward and reverse voltages using the “diode test” function of the DMM (see the Appendix). In this mode the DMM drives a tiny current (0.5 mA) through the diode and measures the voltage difference between the terminals. Record them below:

Forward-biased voltage: 0.584 [V]Reverse-biased voltage: Open [V]

If the forward/reverse voltages are close to zero (or infinite) in both directions, your diode is defective. Don't use “shorted-out” diodes; you may fry the transformer!

Part 1: Half-wave rectifier

Assemble the half-wave rectifier shown in Figure 2. For the time being, leave out the filter capacitor. It will be added later. Use the transformer as the input AC source in the lab. The center-tapped transformer is mounted in a small aluminum box. Do not open the transformer box otherwise you could get a 120 V shock! Connect one of the white terminals and the central (green) terminal to the circuit.

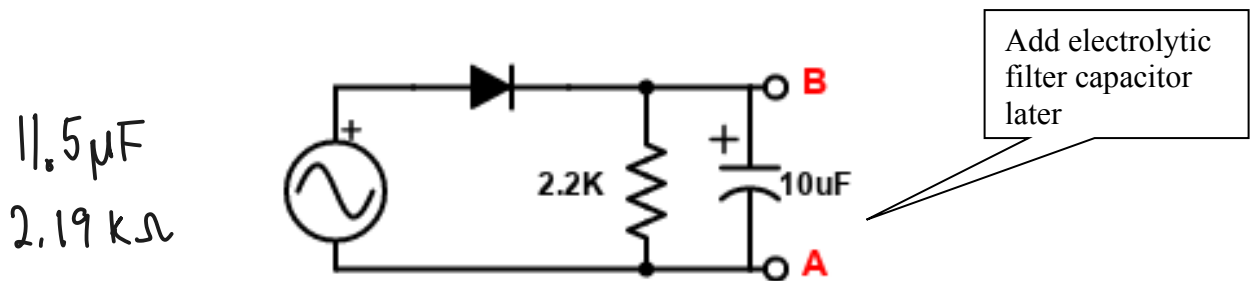


Figure 2. Half-wave rectifier.

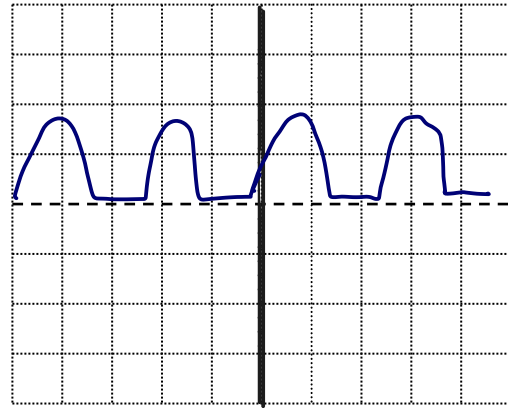
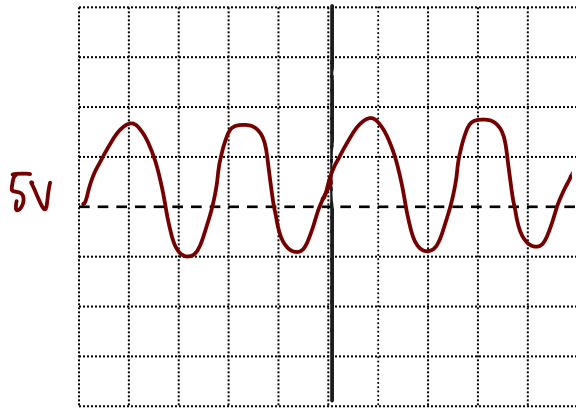
Measure the RMS voltage across the transformer and record its value.

Source RMS voltage: 18.2 [V]

Name: _____

Date: _____

Use the oscilloscope to look at the waveforms at the anode of the diode (the input voltage) and the cathode of the diode (the output voltage) and sketch them, including the scale on each axis.



Questions:

What is the amplitude (NOT peak-to-peak amplitude!) of the input before rectification?

9.04 [V]

What is the maximum voltage after rectification? Why is this not the same as the amplitude before rectification?

8.80 [V]

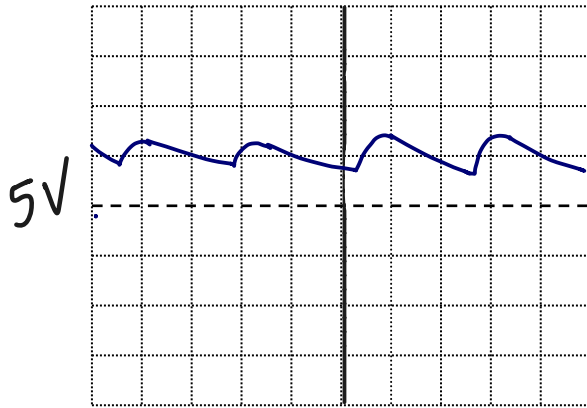
How long is the near-zero “flat region” of the voltage after the diode? Why is it this long?

8.48 [ms]

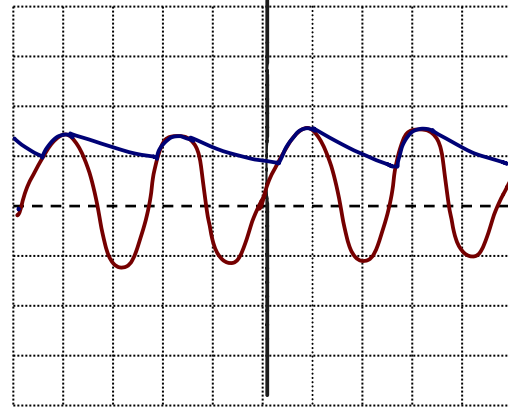
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To smooth out the output, add a 10 μF electrolytic capacitor to the circuit (see Figure 2) and repeat the measurements. Observe the polarity of the electrolytic capacitor! You will find that the "ripple" on the output is greatly reduced, making the power supply more suitable for applications. Use the oscilloscope to look at the filtered output and sketch the waveforms before and after the diode.



4ms



4ms

The peak-to-peak ripple voltage should be approximately: $\Delta V = I_{load} / (f C)$ where $I_{load} = V_{max} / R$ and $f = 60 \text{ Hz}$ (in the US). What is the origin of this formula? Discuss in your write-up. (Think of what fraction of the voltage decays away during one period of the cycle, for a given RC time.) Although this formula is a very rough approximation, it does give the right scaling with the capacitance. Try using a larger capacitor (e.g. 100 μF) and see if your results change in the way this approximation predicts.

Calculate the load current: $I_{load} = \underline{0.00411} \text{ [A]}$

Calculate the ripple voltage: $\Delta V = \underline{5.96} \text{ [V]}$

Measure the ripple voltage: $\Delta V = \underline{4} \text{ [V]}$

$$I_{load} =$$

$$f = 60 \text{ Hz}$$

$$\Delta t = 16.9 \text{ ms}$$

$$V_{max} = 9 \text{ V}$$

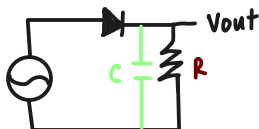
$$R = 2.19 \text{ k}\Omega$$

$$C = 11.5 \mu\text{F}$$

$$I_{load} = V_{max} / R = 9 / 2.19 \text{ k}\Omega = 0.00411 \text{ A}$$

$$\Delta V = \frac{I_{load}}{fC} = \frac{0.00411}{(60 \cdot 11.5 \cdot 10^{-6})} = 5.96 \text{ V}$$

Half-wave
Rectifier
w/ filter (C)



origin:

when discharging

$$V_{out} = (V_{in} - 0.6) e^{-t/RC} = V_{max} (1 - \frac{t}{RC})$$

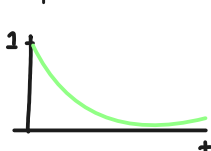
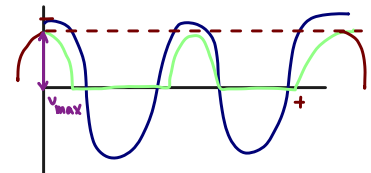
$$\text{for } R \gg T = \frac{2\pi}{\omega}, \quad e^{-t/RC} \approx (1 - \frac{t}{RC})$$

$\Delta V = \text{Ripple} = \text{peak-to-peak}$

← this is the key relation

when C is charging
→ OA to R (discharge path)

$$\Delta V = V_{max} \left(\frac{\Delta t}{RC} \right) = V_{max} \frac{2\pi}{\omega RC} = I_{max} R \frac{2\pi}{\omega RC} = \frac{I_{max}}{fC} \quad (f = \text{frequency})$$



Name: _____

Date: _____

Part 2: Full-wave rectifier

The full-wave rectifier circuit is shown below. When you put it together, pay attention to the direction of the four diodes (otherwise disaster strikes!). Use the transformer as the input AC source in the lab.

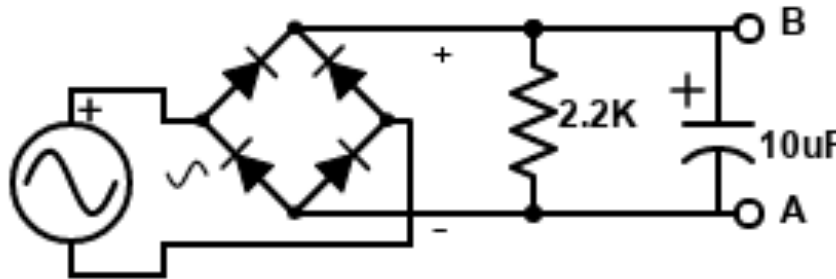
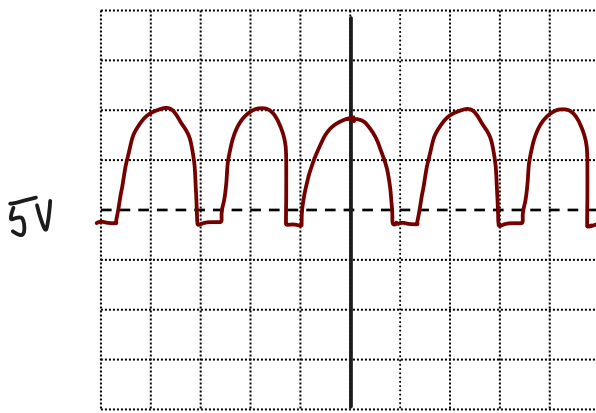


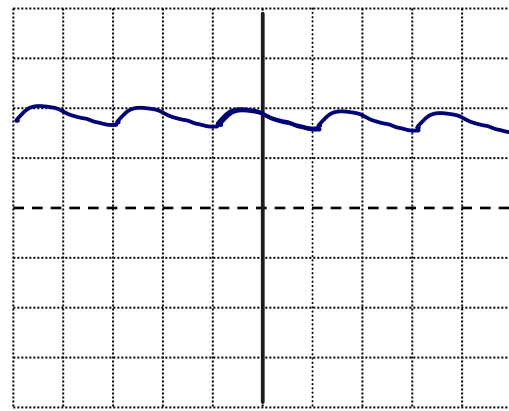
Figure 3. Full-wave rectifier.

Look at the output (V_{ab}) on the oscilloscope without the filter capacitor and then with a $10\ \mu\text{F}$ filter capacitor in place. Sketch the results (don't forget to indicate the scales).

Warning: Do not connect the input and output of the rectifier to the scope at the same time in the lab! Why?



Without filter capacitor.



With filter capacitor

Ripple voltage for full-wave rectifier: $\Delta V = I_{load} / (2fC)$, with $f = 60\ \text{Hz}$ in US. Why is this half of the ripple of a half-wave rectifier? Discuss in your write-up.

Calculate the load current: $I_{load} = \underline{4.11\text{E-3}}\ \text{[A]}$

Calculate the ripple voltage: $\Delta V = \underline{2.98}\ \text{[V]}$

Measure the ripple voltage: $\Delta V = \underline{2}\ \text{[V]}$

$$I_L = \frac{V_{max}}{R} = \frac{9V}{2190\ \Omega} = 4.11\ \text{mA}$$

$$\Delta V = \frac{I_L}{2fC} = \frac{4.11\text{E-3}}{(2 \cdot 60\text{Hz} \cdot 11.5\text{E-6F})} \Rightarrow$$

Name: _____

Date: _____

Part 3: Diode clamping, clipping, limiting

Diode circuits are often used to “clamp” or “clip” voltage signals. In the circuit in Figure 4, the diode starts conducting when the input voltage exceeds the voltage of the power supply (5 V in the figure). The output wave form looks like the input, except that the parts above 5 Volts are “clipped off”. The clipping may not be perfect because the power supply has a non-zero internal resistance and the diode has a finite forward conductance.

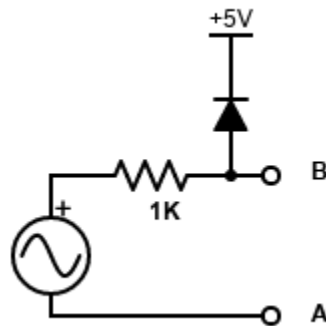
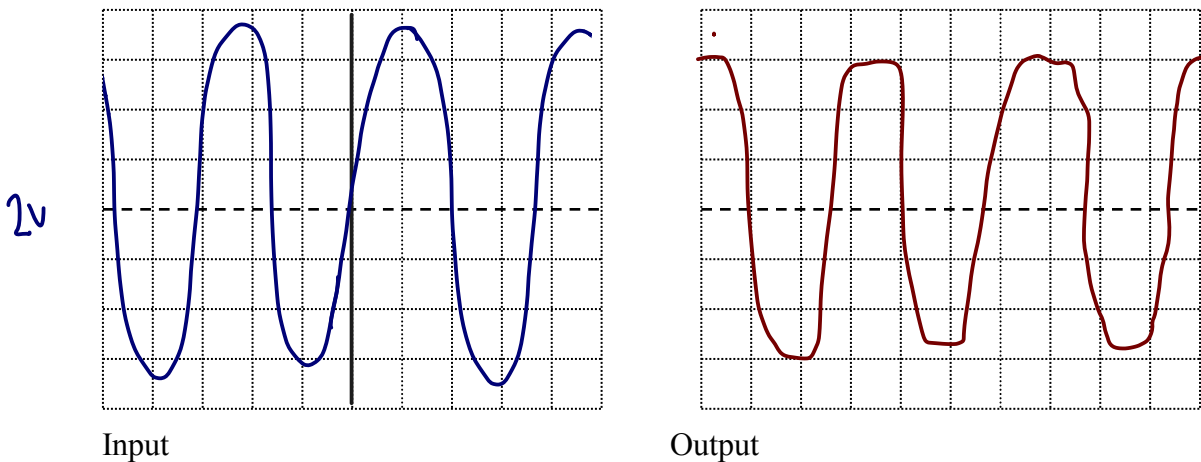


Figure 4. Diode clamping circuit.

Drive the circuit with a sine wave (from the signal generator) and gradually increase the amplitude until it exceeds the power supply voltage. Sketch the input and output wave forms (don’t forget to indicate the scales):



Name: _____

Date: _____

A “diode limiter” is often used to protect sensitive instruments against damaging over-voltages. Assemble the circuit below, use a (symmetric) triangle wave from the signal generator as the input and gradually increase the amplitude starting at 100 mVpp. At a certain amplitude the circuit starts to clip the output waveform.

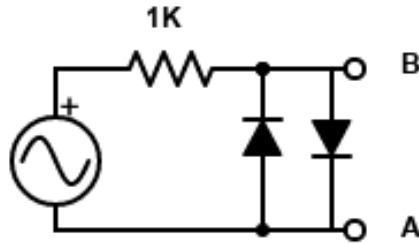
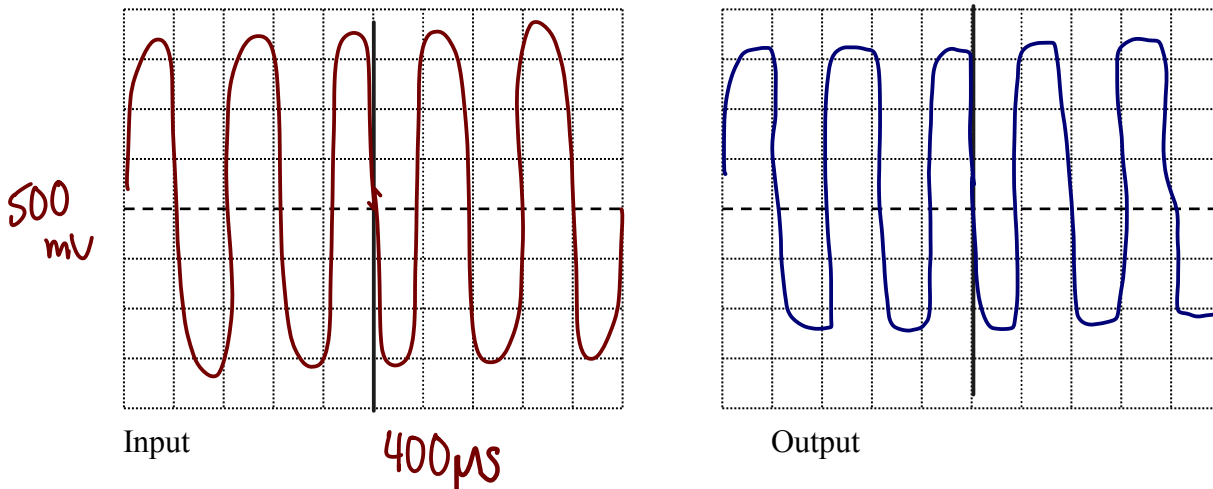


Figure 5. Diode limiter circuit.

Sketch the input and output with the clipping clearly visible.



At what voltage is the output clipped? Measure it on the oscilloscope and explain.

$$V = \pm 0.57 \text{ [V]}$$

Clamps the input down by $V = \pm 0.57V$ alternating
in mag. w/ waveform

Dang

Appendix

Checking diodes

- **Measurement Method:** Uses $0.5\text{ mA} \pm 0.2\%$ constant current source
- **Response Time:** 190 samples/ second with audible tone
- **Input Protection:** 1000 V (HI terminal)

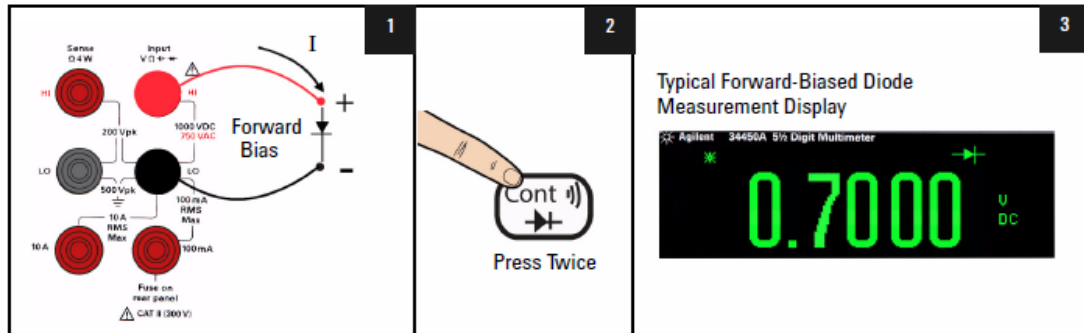


Figure 1-15 Forward-biased diode terminal connection and display

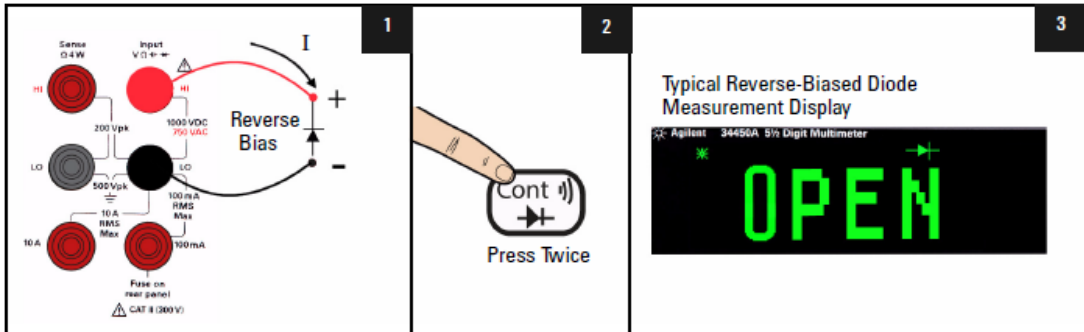


Figure 1-16 Reverse-biased diode terminal connection and display

Figure A.1. How to test the diode using the Agilent 34450A DMM.